



NUST CHIP DESIGN CENTRE

Digital Logic Design

Lab Manual # 11

Communication Protocols (UART using SV)

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1. In-Lab Tasks: (Write your lab task & screenshots here)

The screenshot displays a Verilog IDE interface. A dialog box titled "Bitstream Generation Completed" is open, indicating that the bitstream generation was successful. The dialog includes a "Next" section with four radio button options: "Open Implemented Design" (selected), "View Reports", "Open Hardware Manager", and "Generate Memory Configuration File". There is also a checkbox for "Don't show this dialog again" and "OK" and "Cancel" buttons.

Below the dialog, the "RC Report" dashboard is visible. It shows project details for a Nexys-A7-100T-Master.xdc project. The dashboard is divided into several sections:

- Synthesis:** Status is "Complete" with 1 warning. Messages include "xc7a100tcs324-1".
- Implementation:** Status is "Complete" with 1 warning. Messages include "xc7a100tcs324-1".
- DRC Violations:** Summary shows 1 warning. A link to "Implemented DRC Report" is provided.
- Timing:** Summary shows "Worst Negative Slack (WNS): 6.023 ns", "Total Negative Slack (TNS): 0 ns", "Number of Failing Endpoints: 0", and "Total Number of Endpoints: 28". A link to "Implemented Timing Report" is provided.
- Utilization:** A table showing resource utilization for LUT, FF, IO, and BUFG.
- Power:** A bar chart showing power consumption for Dynamic (0.003 W, 3%), Static (0.097 W, 97%), and PL Static (0.097 W, 100%) components.

Resource	Utilization	Available	Utilization %
LUT	112	63400	0.18
FF	133	126800	0.10
IO	25	210	11.90
BUFG	3	32	9.38

Component	Power (W)	Percentage (%)
Dynamic	0.003	3%
Static	0.097	97%
PL Static	0.097	100%

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